

17.1 Petroleum: a fossil fuel

The fossil fuels

The **fossil fuels** are **petroleum** (or crude oil), **coal**, and **natural gas**. They are called fossil fuels because they are the remains of plants and animals that lived millions of years ago.



Petroleum formed from the remains of dead organisms that fell to the ocean floor, and were buried under thick sediment. High pressures slowly converted them to petroleum, over millions of years.



Natural gas is mainly **methane**. It is often found with petroleum. It is formed in the same way. But high temperatures and high pressures caused the compounds to break down to gas.

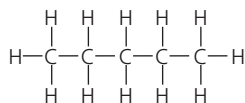


Coal is the remains of lush vegetation that grew in ancient swamps. The dead vegetation was buried under thick sediment. Pressure and heat slowly converted it to coal, over millions of years.

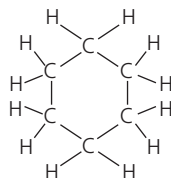
What is in petroleum?

Petroleum is a smelly mixture of hundreds of different compounds. They are **organic compounds**, which means they contain carbon, and usually hydrogen.

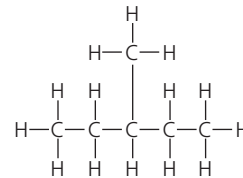
In fact most are **hydrocarbons** – they contain *only* carbon and hydrogen. These drawings show molecules of three different hydrocarbons:



This is a molecule of pentane, C_5H_{12} . It has a straight chain of 5 carbon atoms.



This is a molecule of cyclohexane, C_6H_{12} . Here a chain of 6 carbon atoms form a ring.



This is a molecule of 3-methyl pentane, C_6H_{14} . Here 6 carbon atoms form a **branched** chain.

A formula drawn out in this way is called a **structural formula**.

Notice how the carbon atoms are bonded to each other, to make the spine of each molecule. The hydrogen atoms are bonded to the carbon atoms. In petroleum you will find hydrocarbon molecules of different shapes and sizes, with different numbers of carbon atoms, from 1 to over 70.

How we use petroleum

Over 13 billion litres of petroleum are used around the world every day.



Around half the petroleum pumped from oil wells is used for transport. It provides the fuel for cars, trucks, planes, and ships. You won't get far without it!



Most of the rest is burned for heat, in factories, homes, and power stations, as above. In a power station, the heat is used to turn water to steam, to drive turbines.



A small % is used as the starting chemicals to make many other things: plastics, shampoo, paint, thread, fabric, detergents, makeup, medical drugs, and more.

Many of the things you use every day were probably made from petroleum. Toothbrush, comb, and shampoo just for a start!

A non-renewable resource

Petroleum is still forming, very slowly, under the oceans. But we are using it up much faster than it can form, which means it will run out one day.

So petroleum is called a **non-renewable resource**.

It is hard to tell when it will run out. At the present rate of use, some experts say the world's reserves will last about 40 more years. What will we do then?



▲ A platform for pumping petroleum from under the ocean.

Q

- 1 The other name for petroleum is ... ?
- 2 Why is petroleum called a *fossil fuel*?
- 3 What is a *hydrocarbon*?
- 4 What is petroleum made of?
- 5 Explain why petroleum is such a valuable resource.
- 6 Petroleum is called a *non-renewable* resource. Why?
- 7 What do you think we will use for fuel, when petroleum runs out?

17.2 Refining petroleum

What does *refining* mean?

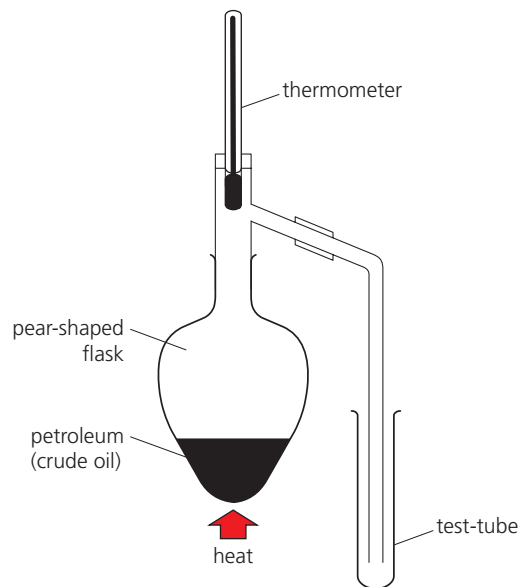
Petroleum contains *hundreds* of different hydrocarbons. But a big mixture like this is not very useful.

So the first step is to separate the compounds into groups with molecules of a similar size. This is called **refining** the petroleum. It is carried out by **fractional distillation**.

Refining petroleum in the lab

The apparatus on the right can be used to refine petroleum in the lab.

- 1 As you heat the petroleum, the compounds start to evaporate. The ones with smaller lighter molecules go first, since it takes less energy to free these from the liquid.
- 2 As the hot vapours rise, so does the thermometer reading. The vapours condense in the cool test-tube.
- 3 When the thermometer reading reaches 100 °C, replace the first test-tube with an empty one. The liquid in the first test-tube is your first **fraction** from the distillation.
- 4 Collect three further fractions in the same way, replacing the test-tube at 150 °C, 200 °C, and 300 °C.



Comparing the fractions

Now compare the fractions – how runny they are, how easily they burn, and so on. You can burn samples on a watch glass, like this:

fraction 1



It catches fire easily. The flame burns high, which shows that the liquid is **volatile** – it evaporates easily.

fraction 2



This catches fire quite easily. The flame burns less high – so this fraction is less volatile than fraction 1.

fraction 3



This seems less volatile than fraction 2. It does not catch fire so readily or burn so easily – it is not so **flammable**.

fraction 4



This one does not ignite easily. You need to use a wick to keep it burning. It is the least flammable of the four.

This table summarizes the results:

Fraction	Boiling point range	How easily does it flow?	How volatile is it?	How easily does it burn?	Size of molecules
1	up to 100 °C	very runny	volatile	very easily	
2	100 – 150 °C	runny	less volatile	easily	
3	150 – 200 °C	not very runny	even less volatile	not easily	
4	200 – 300 °C	viscous (thick and sticky)	least volatile	only with a wick	

The trends the fractions show

Those results show that, the larger the molecules in a hydrocarbon:

- the higher its boiling point will be
- the less volatile it will be
- the less easily it will flow (or the more viscous it will be)
- the less easily it will burn.

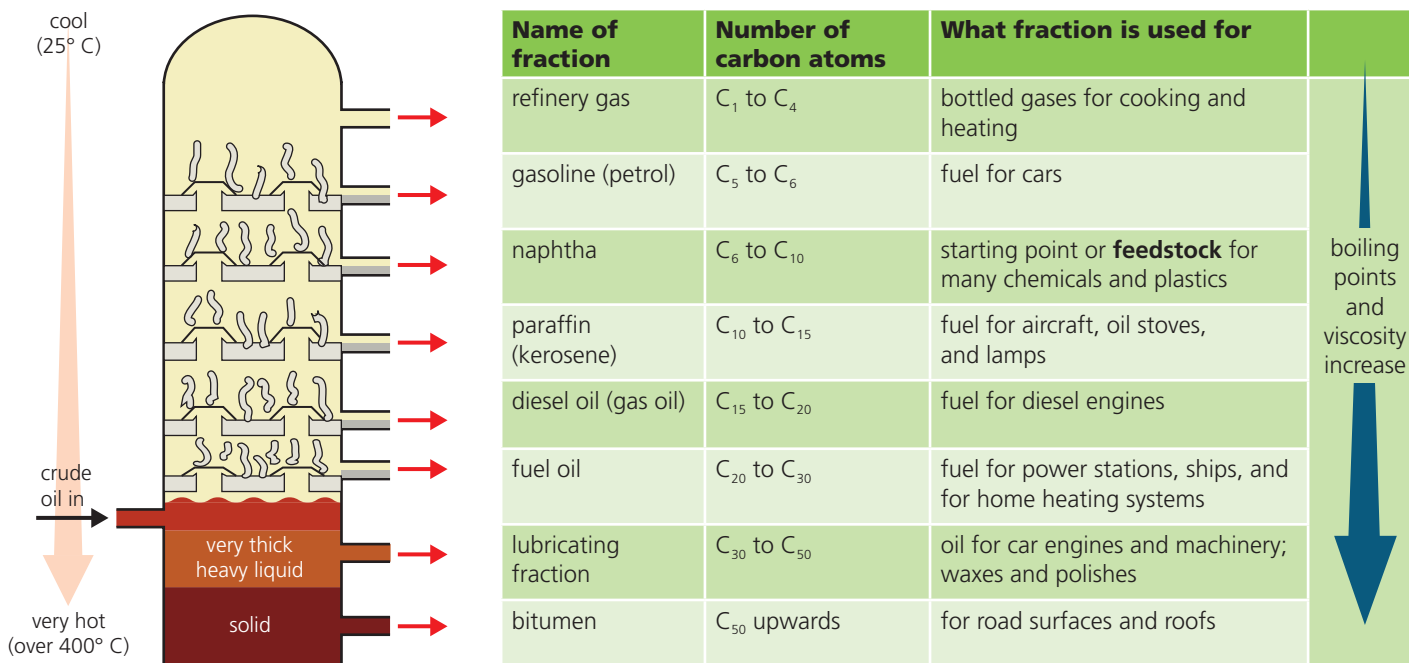
These trends help to dictate what the different fractions will be used for, as you will see below.

In the petroleum refinery

In a refinery, the fractional distillation is carried out in a tower that is kept very hot at the base, and cooler towards the top. Look at the drawing.

Petroleum is pumped in at the base. The compounds start to boil off. Those with the smallest molecules boil off first, and rise to the top of the tower. Others rise only part of the way, depending on their boiling points, and then condense.

The table shows the fractions that are collected.



As the molecules get larger, the fractions get less runny, or more viscous: from gas at the top of the tower to solid at the bottom. They also get less flammable. So the last two fractions in the table are not used as fuels.



▲ The new road surface is bitumen mixed with fine gravel.

Q

- Which two opposite processes take place, during fractional distillation?
- A group of compounds collected during fractional distillation is called a?
- What does it mean? **a** volatile **b** viscous
- List four ways in which the properties of different fractions differ.
- Name the petroleum fraction that: **a** is used for petrol
b has the smallest molecules **c** is the most viscous
d has molecules with 20 to 30 carbon atoms

17.3 Cracking hydrocarbons

After fractional distillation ...

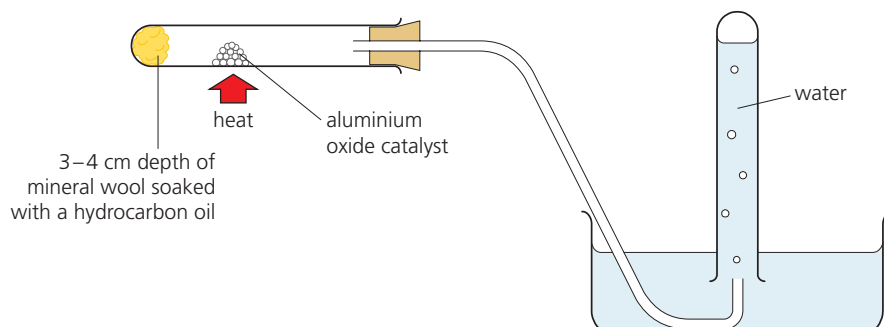
Petroleum is separated into fractions by fractional distillation. But that is not the end of the story. The fractions all need further treatment before they can be used.

- 1 They contain impurities – mainly sulfur compounds. If left in the fuels, these will burn to form harmful sulfur dioxide gas.
- 2 Some fractions are separated further into single compounds, or smaller groups of compounds. For example the gas fraction is separated into methane, ethane, propane, and butane. (We buy butane in canisters.)
- 3 Part of a fraction may be **cracked**.

Cracking breaks molecules down into smaller ones.

Cracking a hydrocarbon in the lab

This experiment is carried out using a hydrocarbon oil from petroleum. The product is a gas, collected over water in the inverted test-tube:



The moment heating is stopped, the delivery tube must be lifted out of the water. Otherwise water will get sucked up into the hot test-tube.

Now compare the reactant and product:

	The reactant	The product
Appearance	thick colourless liquid	colourless gas
Smell	no smell	pungent smell
Flammability	difficult to burn	burns readily
Reactions	few chemical reactions	many chemical reactions

So the product is quite different from the reactant. Heating has caused the hydrocarbon to break down. A **thermal decomposition** has taken place. Note that:

- the reactant had a high boiling point and was not flammable – which means it had large molecules, with long chains of carbon atoms.
- the product has a low boiling point and is very volatile – so it must have small molecules, with short carbon chains.
- the product must also be a hydrocarbon, since nothing new was added.

So the molecules of the starting hydrocarbon have been cracked. And since the product is reactive, it could be a useful chemical.



▲ This sulfur was obtained from sulfur compounds removed from natural gas.



▲ Some of the naphtha fraction from refining will be piped to the cracking plant.

Cracking in the refinery

In the refinery, cracking is carried out in a similar way.

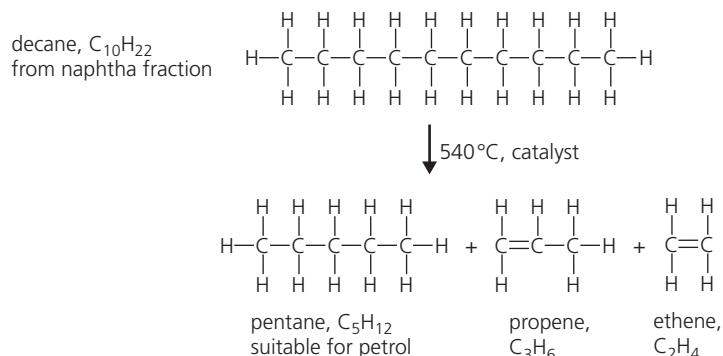
- The long-chain hydrocarbon is heated to vaporize it.
- The vapour is usually passed over a hot catalyst.
- Thermal decomposition takes place.

Why cracking is important

- Cracking helps you make the best use of petroleum. Suppose you have too much of the naphtha fraction, and too little of the gasoline fraction. You can crack some naphtha to get molecules the right size for petrol.
- Cracking *always* produces short-chain compounds with a carbon-carbon double bond. This bond makes the compounds **reactive**. So they can be used to make plastics and other substances.

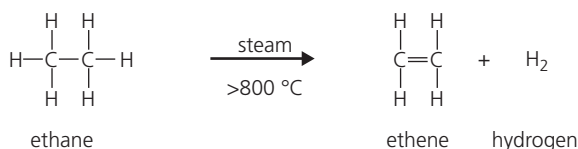
Examples of cracking

- 1 Cracking the naphtha fraction** Compounds in the naphtha fraction are often cracked, since this fraction is used as the feedstock for making many useful chemicals. This is the kind of reaction that occurs:



So decane has been broken down into three smaller molecules. The propene and ethene molecules have carbon-carbon double bonds. These two compounds belong to the **alkene** family, and they are very reactive.

- 2 Cracking ethane** Ethane has very short molecules – but even it can be cracked, to give ethene and hydrogen:



The hydrogen can be used to make ammonia – see page 226.



▲ Plastic furniture: it all began with cracking.



▲ Decane is one of the hydrocarbons in **white spirit**, a solvent used to thin oil-based paint, and clean paintbrushes.

Q

- 1 What happens during cracking?
- 2 Cracking is a *thermal decomposition*. Explain why.
- 3 Describe the usual conditions needed for cracking a hydrocarbon in the petroleum refinery.
- 4 What is *always* produced in a cracking reaction?
- 5 Explain why cracking is so important.
- 6 a A straight-chain hydrocarbon has the formula C_5H_{12} . Draw the structural formula for its molecules.
b Now show what might happen when the compound is cracked.

17.4 Families of organic compounds

What their names tell you

There are *millions* of organic compounds. That could make organic chemistry confusing – but to avoid this, the compounds are named in a very logical way.

The rest of this chapter is about some families of organic compounds.

For these families, the name of the organic compound tells you:

- which family it belongs to
- how many carbon atoms are in it.

Look at these two tables:

If the name ends in ...	... the compound belongs to this family ...	Example
-ane	the alkanes	ethane, C_2H_6
-ene	the alkenes	ethene, C_2H_4
-ol	the alcohols	ethanol, C_2H_5OH
-oic acid	the carboxylic acids	ethanoic acid, CH_3COOH

This in the name ...	... means this many carbon atoms ...	Example from the alkane family
meth-	1	methane, CH_4
eth-	2	ethane, C_2H_6
prop-	3	propane, C_3H_8
but-	4	butane, C_4H_{10}
pent-	5	pentane, C_5H_{12}
hex-	6	hexane, C_6H_{14}



▲ Natural gas burning at a cooker hob: it is mainly methane, the simplest alkane.

The alkanes: the simplest family

Here again are the first four members of the alkane family. Note that methane is the simplest member. What patterns do you notice?

Compound	methane	ethane	propane	butane
Formula	CH_4	C_2H_6	C_3H_8	C_4H_{10}
Structural formula	$\begin{array}{c} H \\ \\ H-C-H \\ \\ H \end{array}$	$\begin{array}{cc} H & H \\ & \\ H-C & -C-H \\ & \\ H & H \end{array}$	$\begin{array}{ccc} H & H & H \\ & & \\ H-C & -C & -C-H \\ & & \\ H & H & H \end{array}$	$\begin{array}{cccc} H & H & H & H \\ & & & \\ H-C & -C & -C & -C-H \\ & & & \\ H & H & H & H \end{array}$
Number of carbon atoms in the chain	1	2	3	4
Boiling point/ $^{\circ}C$	-164	-187	-42	-0.5

boiling point increases with chain length

Comparing families

This table shows one member from each of the four families. Compare them.

Family	A member	Structural formula	Comments
alkanes	ethane, C_2H_6		- The alkanes contain only carbon and hydrogen, so they are hydrocarbons. - The bonds between their carbon atoms are all single bonds.
alkenes	ethene, C_2H_4		- The alkenes are hydrocarbons. - All alkenes contain carbon – carbon double bonds. - The $C=C$ bond is called their functional group.
alcohols	ethanol, C_2H_5OH		- The alcohols are not hydrocarbons. - They are like the alkanes, but with an OH group. - The OH group is their functional group.
carboxylic acids	ethanoic acid, CH_3COOH		- The carboxylic acids are not hydrocarbons. - All carboxylic acids contain the COOH group. - The COOH group is their functional group.

Functional groups

A **functional group** is the part of a molecule that largely dictates how the molecule will react.

For example, all the alkenes have similar reactions because they all have the same functional group, the $C=C$ bond.

Homologous series

Look back at the alkanes in the table at the bottom of page 250.

They form a **homologous series**. In a homologous series:

- All the compounds fit the same general formula.
For the alkanes the general formula is C_nH_{2n+2} , where n is a number.
For methane n is 1, giving the formula CH_4 .
For ethane n is 2, giving C_2H_6 .
For propane n is 3, giving C_3H_8 .
- The chain length increases by 1 each time.
- As the chain gets longer, the compounds show a gradual change in properties. For example, their boiling points rise, and they burn less easily.

As you will see later, all four families in this unit form homologous series.

In a homologous series ...

As the chain gets longer:

- melting and boiling points rise
- viscosity increases – the compounds flow less easily
- flammability decreases – the compounds burn less easily.

Q

1 Propanol is an organic compound.

- How many carbon atoms does it contain?
- Which family does it belong to?
- See if you can draw a structural formula for it.

2 Draw a structural formula for the alkane called hexane.

3 An alkane has 32 carbon atoms in each molecule. Give its formula.

4 Try to draw the structural formula for propanoic acid.

17.5 The alkanes

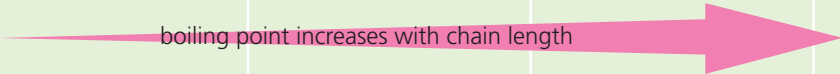
Alkanes: a reminder

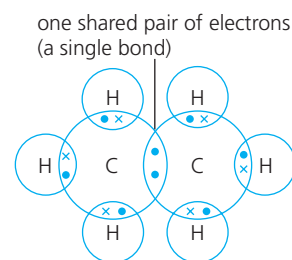
This is what you have learned about the alkanes so far:

- The alkanes are the simplest family of organic compounds.
- They are hydrocarbons: they contain only carbon and hydrogen.
- Their carbon – carbon bonds are all single bonds.
- They form a homologous series, with the general formula C_nH_{2n+2} .

This table shows the first four members of the alkane family. What patterns do you notice?

Compound	methane	ethane	propane	butane
Formula	CH_4	C_2H_6	C_3H_8	C_4H_{10}
Structural formula	<pre> H H — C — H H </pre>	<pre> H H H — C — C — H H H </pre>	<pre> H H H H — C — C — C — H H H H </pre>	<pre> H H H H H — C — C — C — C — H H H H H </pre>
Number of carbon atoms in the chain	1	2	3	4
Boiling point/°C	–164	–87	–42	–0.5

boiling point increases with chain length 



▲ The bonding in ethane.

Key points about the alkanes

- 1 They are found in petroleum and natural gas. Petroleum contains alkanes with up to 70 carbon atoms. Natural gas is mainly methane, with small amounts of ethane, propane, butane, and other compounds.
- 2 The first four alkanes are gases at room temperature. The next twelve are liquids. The rest are solids. Boiling points increase with chain length because attraction between the molecules increases – so it takes more energy to separate them.
- 3 Since all their carbon – carbon bonds are single bonds, the alkanes are called **saturated**. Look at the bonding in ethane on the right above.
- 4 Generally, the alkanes are quite unreactive.
- 5 But alkanes do burn well in a good supply of oxygen, forming carbon dioxide and water vapour, and giving out plenty of heat. So they are used as **fuels**. Methane burns the most easily. Like this:

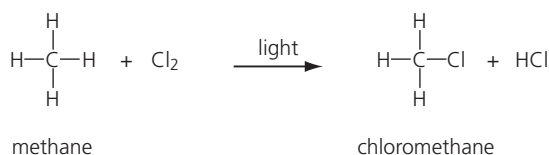
$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) + \text{heat energy}$$
- 6 If there is not enough oxygen, the alkanes undergo **incomplete combustion**, giving poisonous carbon monoxide. For example:

$$2CH_4(g) + 3O_2(g) \longrightarrow 2CO(g) + 4H_2O(l) + \text{less heat energy}$$



▲ Butane is used as fuel for cooking and heating, in many homes.

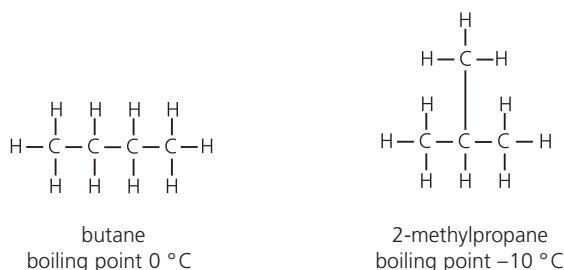
7 Alkanes also react with chlorine in sunlight. For example:



This is called a **substitution** reaction, because a chlorine atom takes the place of a hydrogen atom. If there is enough chlorine, all four hydrogen atoms will be replaced, one by one. Look at the panel on the right.

The reaction can be explosive in sunlight. But it will not take place in the dark, because it is also a **photochemical reaction**: light energy is needed to break the bonds in the chlorine molecules, to start the reaction off.

Isomers



Compare these alkane molecules. Both have the same formula, C₄H₁₀. But they have different structures. The first has a **straight** or unbranched chain. In the second, the chain is **branched**.

The two compounds are **isomers**.

Isomers are compounds with the same formula, but different structures.

The more carbon atoms in a compound, the more isomers it has. There are 75 isomers with the formula C₁₀H₂₂, for example.

Since isomers have different structures, they also have slightly different properties. For example branched isomers have lower boiling points, because the branches make it harder for the molecules to get close. So the attraction between them is less strong, and less heat is needed to overcome it.

Chlorine and methane

The hydrogen atoms can be replaced one by one:

chloromethane	CH ₃ Cl
dichloromethane	CH ₂ Cl ₂
trichloromethane	CHCl ₃
tetrachloromethane	CCl ₄

All four are used as solvents. But they can cause health problems, so are being used less and less.



▲ Branched-chain hydrocarbons burn less easily than their straight-chain isomers. So they are used in petrol to control combustion and stop engine 'knock' – and especially for racing cars.

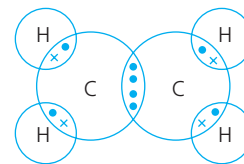
Q

- 1 Describe the bonding in ethane. A drawing will help!
- 2 Why are alkanes such as methane and butane used as fuels? See if you can give *at least* two reasons.
- 3 Butane burns in a similar way to methane. See if you write a balanced equation for its combustion.
- 4 a The reaction of chlorine with methane is called a *substitution* reaction. Why?
b What special condition is needed, for this reaction?
- 5 Ethane reacts with chlorine, in a substitution reaction.
 - a Draw the structural formula for each compound that can form, as the reaction proceeds. (Isomers too!)
 - b Write the formula for each compound in a.
- 6 The compound C₅H₁₂ has *three* isomers.
 - a Draw the structures of these three isomers.
 - b Their boiling points are 9.5, 28, and 36 °C. Match these to your drawings, and explain your choice.

17.6 The alkenes

The alkene family

- The alkenes are hydrocarbons.
- They form a homologous series, with the general formula C_nH_{2n} .
- They all contain the $C=C$ double bond. This is their functional group, and largely dictates their reactions. Look at the bonding in ethene.
- Because they contain $C=C$ double bonds, they are called **unsaturated**. (Alkanes have only single carbon – carbon bonds, so are **saturated**.)



▲ The bonding in ethene.

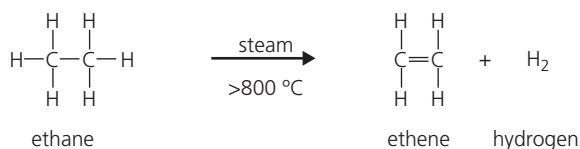
Here are the first three members of the family. Note how ethene is drawn:

Compound	ethene	propene	but-1-ene
Formula	C_2H_4	C_3H_6	C_4H_8
Structural formula	$\begin{array}{c} H & & H \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ H & & H \end{array}$	$\begin{array}{c} H & & H & & H \\ & \diagdown & / & & \\ & C = C & - & C - H \\ & / & & \\ H & & H & & H \end{array}$	$\begin{array}{c} H & & H & & H & & H \\ & \diagdown & / & & & & \\ & C = C & - & C - C - H \\ & / & & & \\ H & & H & & H & & H \end{array}$
Number of carbon atoms	2	3	4
Boiling point/ $^{\circ}C$	-102	-47	-6.5

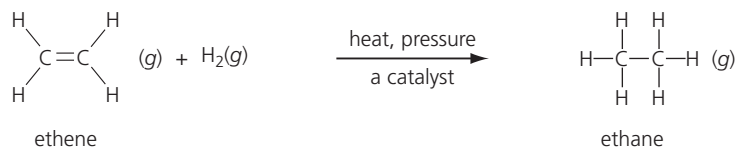
Do you agree that these compounds fit the general formula C_nH_{2n} ?

Key points about the alkenes

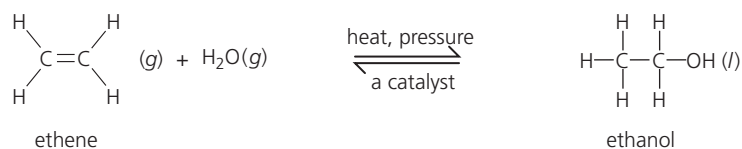
- 1 The alkenes are made from alkanes by cracking. For example ethene is formed by cracking ethane. Hydrogen is also produced:



- 2 Alkenes are much more reactive than alkanes, because the double bond can break, to add on other atoms. For example, ethene can add on hydrogen again, to form ethane:



It also adds on water (as steam) to form ethanol, an alcohol:



These reactions are called **addition reactions**. Can you see why?

An addition reaction turns an unsaturated alkene into a saturated compound.



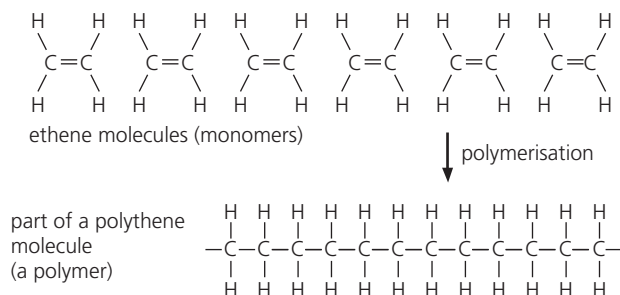
▲ Ethanol is a good solvent. It also kills many germs, so it is used in disinfectant gels for wiping your hands, in hospital.

Polymerisation

Alkene molecules undergo a very useful **addition reaction**, where they add on *to each other* to form compounds with very long carbon chains.

The alkene molecules are called **monomers**. The long-chain compounds that form are called **polymers**. The reaction is called **polymerisation**.

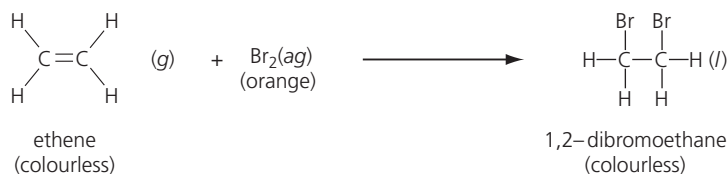
For example ethene polymerises like this:



The product is **poly(ethene)** or **polythene**. The chain can be many thousands of carbon atoms long!

A test for unsaturation

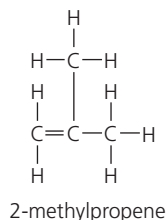
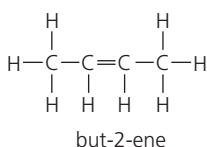
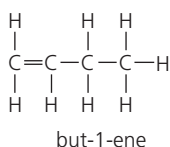
You can use **bromine water** to test whether a hydrocarbon is unsaturated. It is an orange solution of bromine in water. If a C=C bond is present, an addition reaction takes place and the colour disappears. For example:



Isomers in the alkene family

In alkenes, the chains can branch in different ways, *and* the double bonds can be in different positions.

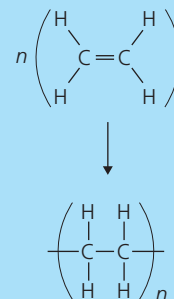
Compare the three compounds below. All three have the formula C₄H₈, but they have different structures. So they are isomers.



Now look at the numbers in their three names. What do these tell you?

The short way to show it ...

You can show polymerisation in a short way, like this. *n* stands for a very large number.



▲ Polythene is used to make plastic bottles, plastic bags, plastic sheeting ...

Q

- 1 a Name the two simplest alkenes.
b Now draw their structural formulae.
- 2 What makes alkenes react so differently from alkanes?
- 3 Ethene can *polymerise*. What does that mean?
- 4 a Propene is *unsaturated*. What does that mean?
b Write an equation for its reaction with bromine.
- 5 How would you turn propene into:
a propane? b propanol?

17.7 The alcohols

What are alcohols?

The **alcohols** are the family of organic compounds that contain the **OH** group. This table shows the first four members:

Alcohol	methanol	ethanol	propan-1-ol	butan-1-ol
Formula	CH_3OH	$\text{C}_2\text{H}_5\text{OH}$	$\text{C}_3\text{H}_7\text{OH}$	$\text{C}_4\text{H}_9\text{OH}$
Structural formula	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$
Number of carbon atoms	1	2	3	4
Boiling point/ $^{\circ}\text{C}$	65	78	87	117

Note that:

- they form a homologous series, with the general formula $\text{C}_n\text{H}_{2n+1}\text{OH}$.
- their OH functional group means they will all react in a similar way.
- two of the names above have -1- in. This tells you that the OH group is attached to a carbon atom at one end of the chain.

Ethanol, an important alcohol

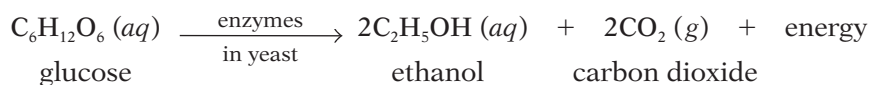
- Ethanol is the alcohol in alcoholic drinks.
- It is a good solvent. It dissolves many substances that do not dissolve in water.
- It evaporates easily – it is **volatile**. That makes it a suitable solvent to use in glues, printing inks, perfumes, and aftershave

Two ways to make ethanol

Ethanol is made in two ways, one biological and one chemical.

1 By fermentation – the biological way

Ethanol is made from glucose using yeast, in the absence of air:



- Yeast is a mass of living cells. The enzymes in it catalyse the reaction. (See page 142.)
- The process is called **fermentation**, and it is exothermic.
- Ethanol can be made in this way from any substance that contains sugar, starch, or cellulose. (These break down to glucose.) For example it can be made from sugarcane, maize, potatoes, and wood.
- The yeast stops working when the % of ethanol reaches a certain level, or if the mixture gets too warm.
- The ethanol is separated from the final mixture by fractional distillation.



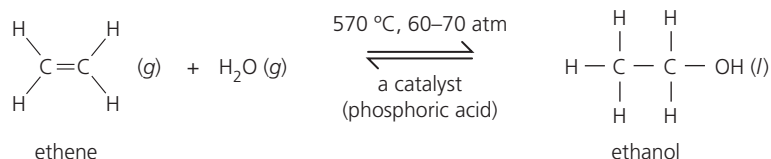
▲ Ethanol is used as a solvent for perfume. Why?



▲ Corn (maize) is widely grown in the USA to make ethanol, for car fuel. Fuel made from plant material, using yeast or bacteria, is called **biofuel**.

2 By the hydration of ethene – the chemical way

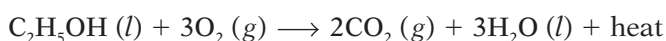
Hydration means water is added on. This is an **addition reaction**.



- The reaction is reversible, and exothermic.
- High pressure and a low temperature would give the best yield. But in practice the reaction is carried out at 570°C , to give a decent rate of reaction.
- A catalyst is also used, to speed up the reaction.

Ethanol as a fuel

Ethanol burns well in oxygen, giving out plenty of heat:



It is increasingly used as a fuel for car engines because:

- it can be made quite cheaply from waste plant material
- many countries have no petroleum of their own, and have to buy it from other countries; it costs a lot, so ethanol is an attractive option
- ethanol has less impact on carbon dioxide levels than fossil fuels do.

Ethanol and global warming

Like the fossil fuels, ethanol does produce carbon dioxide when it burns. This is a greenhouse gas, linked to global warming. But ethanol has less impact on carbon dioxide levels in the atmosphere, because ...



... although carbon dioxide is given out when ethanol burns ...



... it is taken in by plants being grown to make more ethanol.

By contrast, the carbon dioxide given out when fossil fuels burn was taken in from the atmosphere many millions of years ago.



▲ Traffic in Rio de Janeiro in Brazil: running mainly on ethanol made from sugar cane.

The drawbacks

More and more crops are being grown to make ethanol, for cars.

- But that takes up a lot of land.
- It means less land to grow crops for food.
- A shortage of food crops means a rise in food prices.
- When food prices rise, it affects poor people the most.

Many people are against growing crops to make ethanol. They say: *Feed people, not cars!*

Q

- 1 All alcohols react in a similar way. Why?
- 2 Draw the structural formula for ethanol.
- 3 Give three uses of ethanol.
- 4 In Brazil, sugarcane is used to make ethanol. Name the process used, and say what the catalyst is.

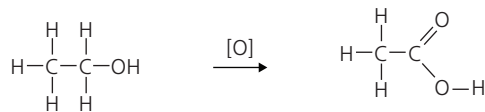
- 5 a Write a word equation for the combustion (burning) of:
 - i ethanol
 - ii methane
 b Compare the equations. What do you notice?
- 6 There is another isomer with the same formula as propan-1-ol. Draw its structure, and suggest a name.

17.8

Here are the first four members of the family:

Name of acid	methanoic	ethanoic	propanoic	butanoic
Formula	HCOOH	CH ₃ COOH	C ₂ H ₅ COOH	C ₃ H ₇ COOH
Structural formula	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C} \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{H} \quad \text{O} \\ \quad \parallel \\ \text{H}-\text{C}-\text{C} \\ \quad \\ \text{H} \quad \text{OH} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{OH} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{OH} \end{array}$
Number of carbon atoms	1	2	3	4
Boiling point/°C	101 °C	118 °C	141 °C	164 °C

- Ethanoic acid is made by oxidising ethanol:

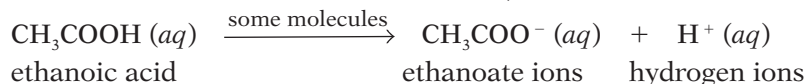


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Ethanoic acid: typical acid reactions

Ethanoic acid shows typical acid reactions.

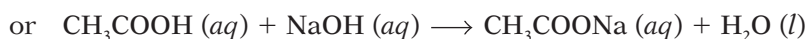
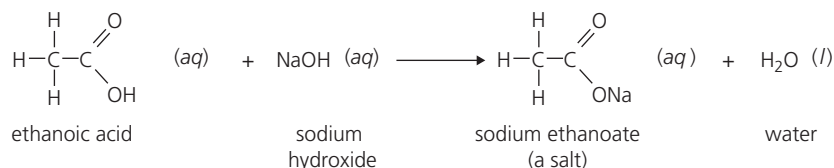
- 1 A solution of ethanoic acid turns litmus red.
- 2 A solution of ethanoic acid contains H^+ ions, because some of the ethanoic acid molecules dissociate in water, like this:



Since only *some* molecules dissociate, ethanoic acid is a **weak acid**.

- 3 Ethanoic acid reacts with metals, bases, and carbonates, to form **salts**.

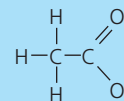
It reacts with sodium hydroxide like this:



Like all salts, sodium ethanoate is an ionic compound.

The ethanoate ion

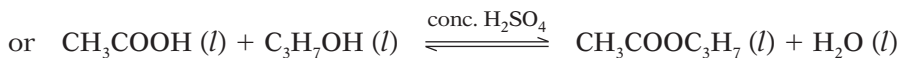
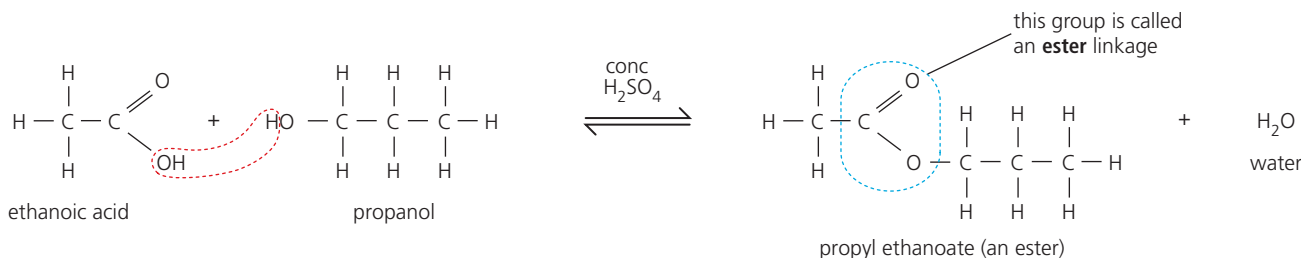
This is its structural formula:



Esters

Ethanoic acid also reacts with alcohols, to give compounds called **esters**.

The alcohol molecule is reversed below, to help you see what is happening:



Note these points:

- Two molecules have joined to make a larger molecule, with the loss of a small molecule, water. So this is called a **condensation reaction**.
- The reaction is reversible, and sulfuric acid acts a catalyst.
- The alcohol part comes *first* in the name – but second in the formula.
- Propyl ethanoate smells of pears. In fact many esters have attractive smells and tastes. So they are added to shampoos and soaps for their smells, and to ice cream and other foods as flavourings.



▲ The smell and taste of the apple come from natural esters. Synthetic esters are used in the shampoo.

Q

- 1 What is the functional group of the carboxylic acids?
- 2 Copy and complete. (Page 152 may help!)
 carboxylic acid + metal $\rightarrow$ _____ + _____
 carboxylic acid + alkali $\rightarrow$ _____ + _____
 carboxylic acid + alcohol $\rightleftharpoons$ _____ + _____
- 3 Carboxylic acids are *weak* acids. Explain why.
- 4 Draw structural formulae to show the reaction between ethanol and ethanoic acid, and name the products.
- 5 What is a *condensation reaction*?
- 6 Esters are important compounds in industry. Why?

Checkup on Chapter 17

Revision checklist

Core curriculum

Make sure you can ...

- ☐ name the fossil fuels and say how they were formed
- ☐ explain what a *hydrocarbon* is
- ☐ explain why petroleum has to be refined, and:
 - describe the refining process
 - name the different fractions
 - say what these fractions are used for
- ☐ explain what *cracking* is, and why it is so useful, and give the equation for the cracking of ethane
- ☐ say what a *structural formula* is, and draw structural formulae for
methane ethane ethene ethanol ethanoic acid
- ☐ say what family a compound belongs to, from its structural formula or its name
- ☐ give the functional groups for the alkenes, alcohols, and carboxylic acids
- ☐ describe alkanes as unreactive except for burning, and give an equation for the combustion of methane
- ☐ explain what *unsaturated* means and describe a test to identify an unsaturated hydrocarbon
- ☐ describe how ethene monomers add on to each other to form the polymer poly(ethene)
- ☐ describe the two ways of making ethanol
- ☐ give at least two uses for ethanol
- ☐ give the reaction for the combustion of ethanol
- ☐ give at least two advantages of ethanol as a fuel

Extended curriculum

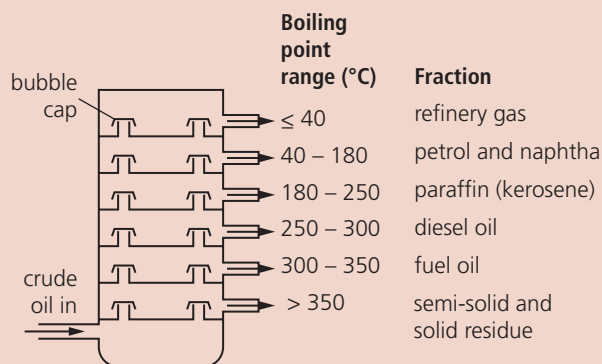
Make sure you can also ...

- ☐ name, and draw the structural formulae for, the four simplest members of each family: alkanes, alkenes, alcohols, and carboxylic acids
- ☐ give the general properties of a homologous series
- ☐ explain what *isomers* are, and draw examples
- ☐ describe the substitution reactions of alkanes with chlorine
- ☐ describe the addition reactions of alkenes with hydrogen and steam
- ☐ describe the two ways to make ethanoic acid
- ☐ explain why ethanoic acid is a weak acid
- ☐ give examples of reactions to show that ethanoic acid is a typical acid
- ☐ describe the reaction of ethanoic acid with an alcohol, and name the products that form

Questions

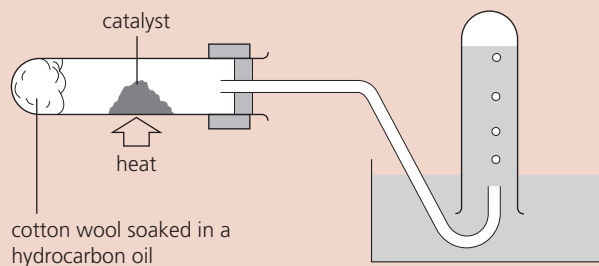
Core curriculum

1 Petroleum is separated into fractions, like this:



- a i What is this process called?
ii It uses the fact that different compounds have different What is missing?
- b i Is naphtha just one compound, or a group of compounds? Explain.
ii Using the terms *evaporation* and *condensation*, explain how naphtha is produced.
- c Give one use for each fraction obtained.
- d A hydrocarbon has a boiling point of 200 °C.
i Are its carbon chains shorter, or longer, than those found in naphtha?
ii Is it more viscous, or less viscous, than the compounds found in naphtha?

2



A hydrocarbon can be cracked in the lab using the apparatus above.

- a What is cracking?
- b Which two things are needed, to crack the hydrocarbon?
- c The first tube of collected gas is discarded. Why? (What else is in the heated tube?)
- d At the end of the experiment, the delivery tube must be removed from the water immediately. Why is this?
- e Ethane, C₂H₆, can be cracked to give ethene, C₂H₄, and hydrogen. Write an equation for this.

- 3** Answer these questions about the alkanes.
- Which two elements do alkanes contain?
 - Which alkane is the main compound in natural gas?
 - After butane, the next two alkanes in the series are *pentane* and *hexane*. How many carbon atoms are there in a molecule of:
 - pentane?
 - hexane?
 - Will pentane react with bromine water? Explain.
 - Alkanes burn in a good supply of oxygen. Name the gases formed when they burn.
 - Write the word equation for the complete combustion of pentane in oxygen.
 - Name a harmful substance formed during *incomplete* combustion of pentane in air.
- 4** When ethanol vapour is passed over heated aluminium oxide, a dehydration reaction occurs, and the gas ethene is produced.
- Draw a diagram of suitable apparatus for carrying out this reaction in the lab.
 - What is meant by a *dehydration reaction*?
 - Write an equation for this reaction, using the structural formulae.
 - What will you see if the gas that forms is bubbled through bromine water?
 - You will not see this if ethanol vapour is passed through bromine water. Why not?
- 5** a Which of these could be used as monomers for addition polymers? Explain your choice.
- ethene, $\text{CH}_2=\text{CH}_2$
 - ethanol, $\text{C}_2\text{H}_5\text{OH}$
 - propane, C_3H_8
 - styrene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$
 - chloropropene, $\text{CH}_3\text{CH}=\text{CHCl}$
- b Suggest a name for each polymer obtained.
- 7** Ethanol is a member of a homologous series.
- Give two general characteristics of a homologous series.
 - Which homologous series is ethanol part of?
 - What is the general formula for the series?
 - What does *functional group* mean?
 - What is the functional group in ethanol's homologous series?
 - Write down the formula of ethanol.
 - Draw the structural formula for the fifth member of the series, pentan-1-ol.
 - Draw the structural formula for an isomer of pentan-1-ol.
 - Describe how pent-1-ene could be made from pentan-1-ol.
 - Name the organic product formed when pentan-1-ol is oxidised using acidified potassium manganate(VII).
- 8** Ethanoic acid is a member of the homologous series with the general formula $\text{C}_n\text{H}_{2n}\text{O}_2$.
- Name this series.
 - What is the functional group of the series?
 - Ethanoic acid is a *weak* acid. Explain what this means, using an equation to help you.
 - Ethanoic acid reacts with carbonates.
 - What would you *see* during this reaction?
 - Write a balanced equation for the reaction with sodium carbonate.
 - Name the member of the series for which $n = 3$, and draw its structural formula.
 - Give the equation for the reaction between this compound and sodium hydroxide.
- 9** Ethanoic acid reacts with ethanol in the presence of concentrated sulfuric acid.
- Name the organic product formed.
 - Which type of compound is it?
 - How could you tell quickly that it had formed?
 - What is the function of the sulfuric acid?
 - The reaction is *reversible*. What does this mean?
 - Write an equation for the reaction.
- 10** Hex-1-ene is an unsaturated hydrocarbon. It melts at -140°C and boils at 63°C . Its empirical formula is CH_2 . Its relative molecular mass is 84.
- To which family does hex-1-ene belong?
 - What is its molecular formula?
 - Hex-1-ene reacts with bromine water. Write an equation to show this reaction.
 - What is this type of reaction called?
 - What would you *see* during the reaction?

Extended curriculum

- 6** The saturated hydrocarbons form a homologous series with the general formula $\text{C}_n\text{H}_{2n+2}$.
- What is a *homologous series*?
 - Explain what the term *saturated* means.
 - Name the series described above.
 - Give the formula and name for a member of this series with two carbon atoms.
 - Draw its structural formula.
 - Name a homologous series of *unsaturated* hydrocarbons, and give its general formula.
 - Give the formula and name for the member of this series with two carbon atoms.
 - Draw the structural formula for the compound.